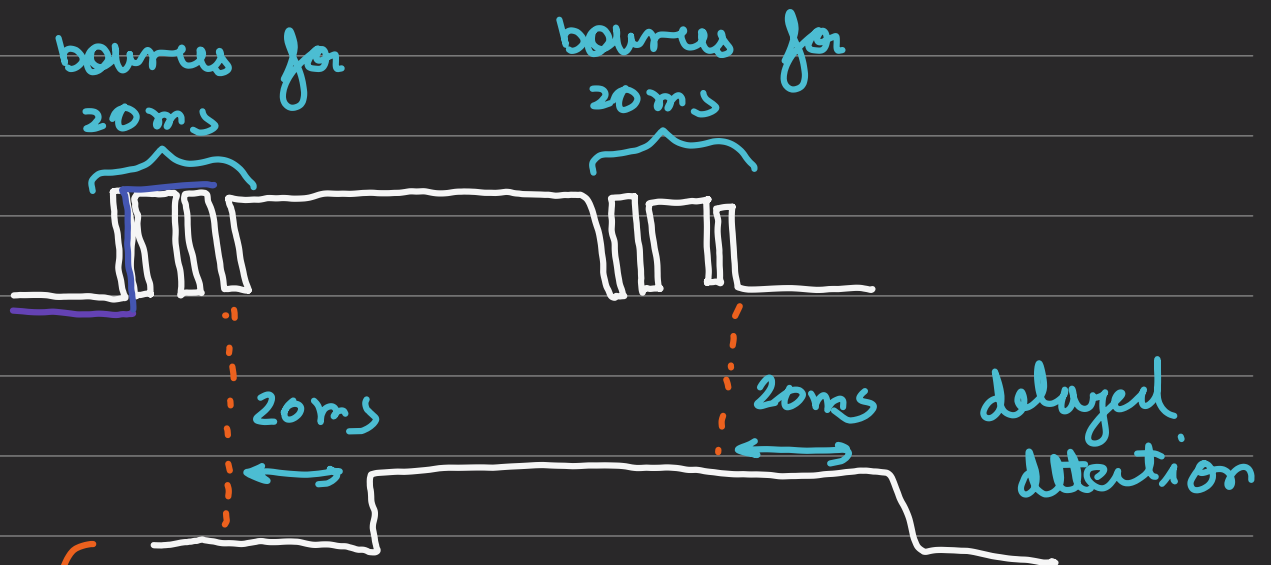
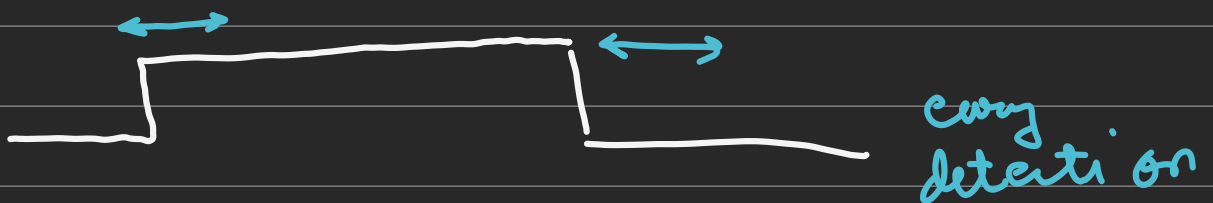


Buttons (Debouncing & Edge Detection)

problem :



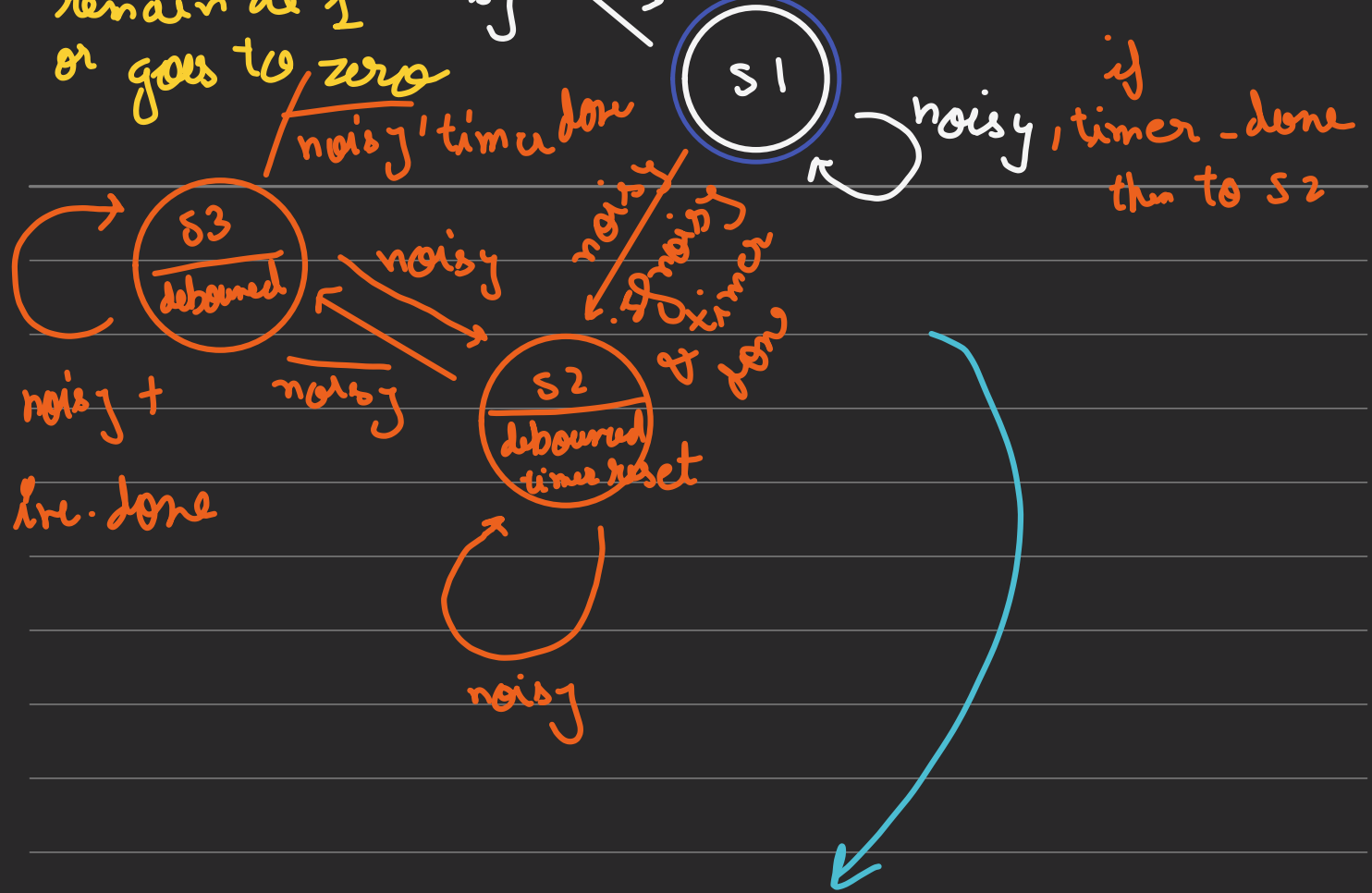
wait for stable position
then wait for specific period



Debouncer FSM



as long as noisy input
of button is at 0, we
will stay in this
state



S0: reset condition, before noise & 0 by default.

S1: when noise after S0.

Starts a timer (timer done)

if noise remains & timer completes
 ↳ custom (20 ms)

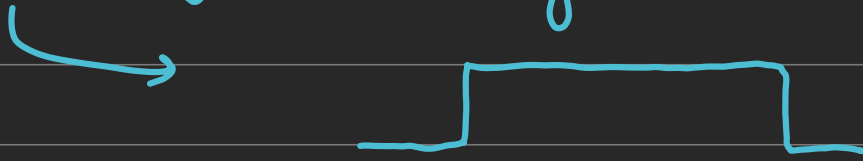
goes to S2 bcs noise + ~ timer

remains at S1.

if noise ends before timer (~

noise) then goes back to S0

S2: if (~noise) goes to S3:



that high portion

else if (noise) remains at S2;

S3: that down portion

if (noise) goes to S2;

if (~noise & ~timer) → S3;
if (~noise & timer) → S0;

↳ timer reset → S0/S2
debounced → S2/S3

→
Go to next slide if
unable to understand

